

Chapter 5

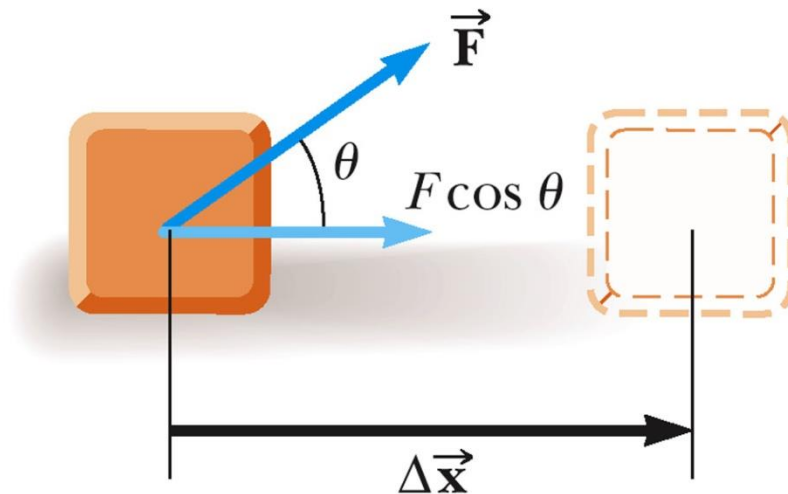
Work, Energy and Power

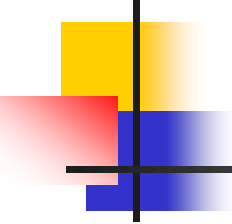
Work

The work, W , done by a constant force on an object is defined as the product of the component of the force along the direction of displacement and the magnitude of the displacement.

Mathematically;

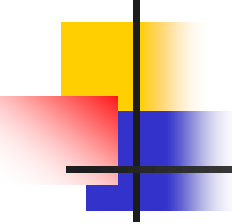
$$W \equiv (F \cos \theta) \Delta x$$





Units of Work

- SI Unit; Newton•meter = Joule
 - $N \cdot m = J$
 - $J = kg \cdot m^2 / s^2$
- If there are multiple forces acting on an object, the total work done is the algebraic sum of the amount of work done by each force
- Work can be done by friction and the energy lost to friction by an object goes into other forms of energy e.g heat



example

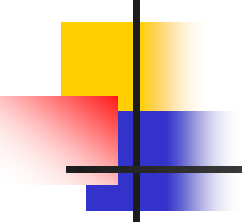
How much work is done on a vacuum cleaner pulled 3 m by a force of 50 N at an angle of 30° above the horizontal?

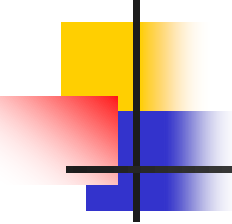
Solution

$$F = 50\text{N} \quad , \quad d = 3\text{m}, \quad \theta = 30^\circ$$

$$W = (F\cos\theta) \times d$$

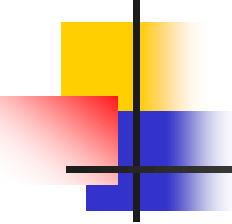
$$\begin{aligned} W &= (50\text{N})(\cos 30^\circ)(3\text{m}) \\ &= 130 \text{ J} \end{aligned}$$

- 
-
- Falling object converts gravitational potential energy into kinetic energy
 - Friction converts kinetic energy into vibrational (thermal) energy
 - makes things hot (rub your hands together)
 - irretrievable energy
 - Doing work *on* something changes that object's energy by amount of



Forms of Energy

- Mechanical
 - Focus for now
 - May be kinetic (associated with motion) or potential (associated with position)
- Chemical
- Electromagnetic
- Nuclear

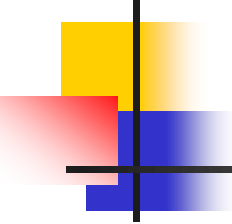


Kinetic Energy:

- * Energy associated with an object in **motion**
- * Depends on speed and mass
- * Scalar quantity
- * SI unit for all forms of energy = **Joule (J)**

$$\text{KE} = \frac{1}{2} mv^2$$

$$\text{KE} = \frac{1}{2} \times \text{mass} \times (\text{velocity})^2$$



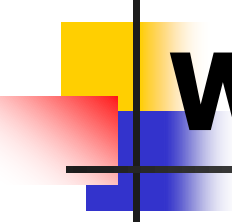
Example

- A 7 kg bowling ball moves at 3 m/s. How fast must a 2.45 g tennis ball move in order to have the same kinetic energy as the bowling ball?

Solution

$$KE_{\text{tennis}} = KE_{\text{bow}}$$

$$\text{Velocity of tennis ball} = 160 \text{ m/s}$$



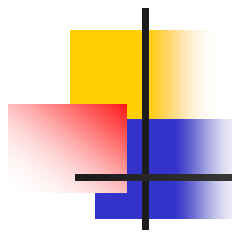
Work-Kinetic Energy Theorem

Work-kinetic Energy Theorem:

- Net work done on a particle equals the change in its kinetic energy (KE)

$$W = \Delta KE$$

$$W = KE_f - KE_o = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_o^2$$



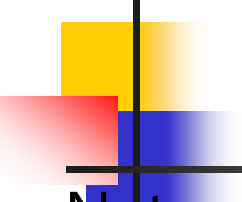
Example

On a frozen pond, a person kicks a 10 kg sled, giving it an initial speed of 2.2 m/s. How far does the sled move if the coefficient of kinetic friction between the sled and the ice is 0.10?

solution

$$m = 10 \text{ kg} \quad v_i = 2.2 \text{ m/s} \quad v_f = 0 \text{ m/s} \quad \mu_k = 0.10$$

$$d = ?$$


$$W_{\text{net}} = F_{\text{net}} d \cos \theta$$

- Net work done of the sled is provided by the force of kinetic friction; $W_{\text{net}} = F_k d \cos \theta \rightarrow F_k = \mu_k N \rightarrow N = mg$

$$W_{\text{net}} = \mu_k mg d \cos \theta$$

- The force of kinetic friction is in the direction opposite of $d \rightarrow \theta = 180^\circ$

- Sled comes to rest $\rightarrow$ So, final KE = 0

$$W_{\text{net}} = \Delta KE = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2$$

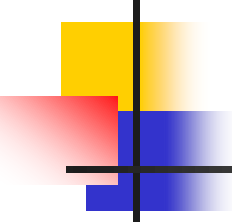
$$W_{\text{net}} = -\frac{1}{2} mv_i^2$$

Use the work-kinetic energy theorem, and solve for d

$$W_{\text{net}} = \Delta KE$$

$$-\frac{1}{2} mv_i^2 = \mu_k mg d \cos \theta$$

$$d = 2.5 \text{ m}$$



Potential Energy:

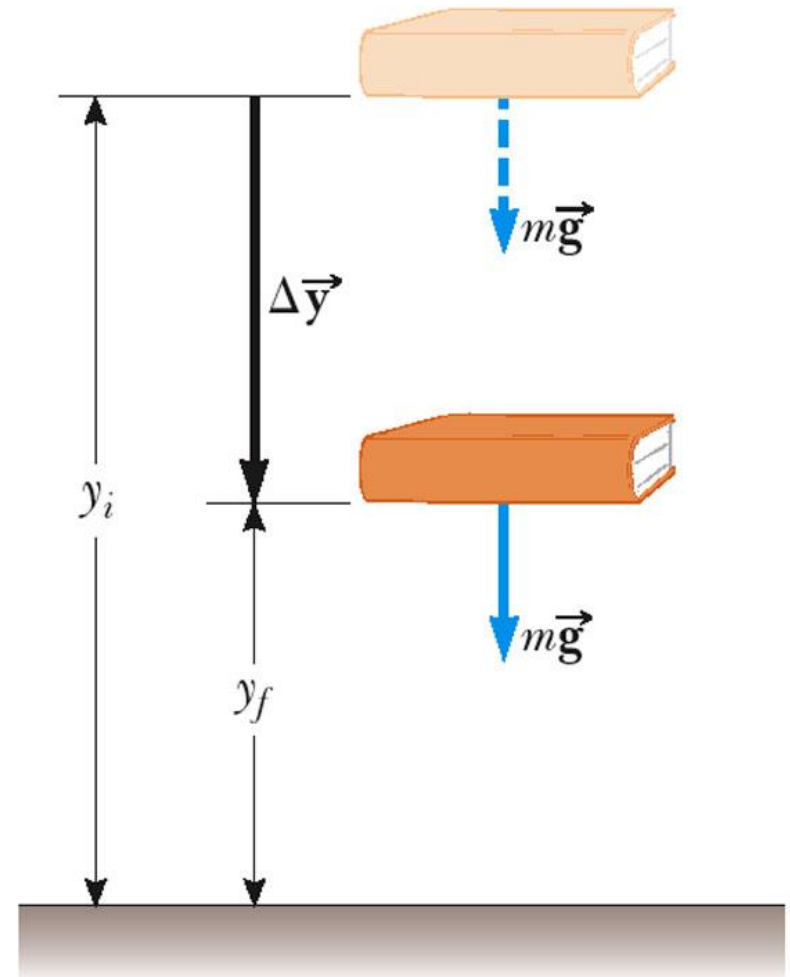
- * Stored energy

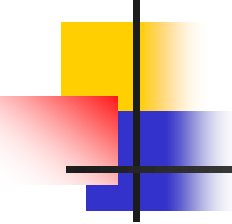
- * Associated with an object that has the potential to move because of its position relative to some other location

- Gravitational potential energy PE_g
- **$PE_g = mgh$** *Where m is mass of an object at a height of h from Earth's surface*
- **SI Unit = Joule (J)**

Work and Gravitational Potential Energy

- $PE = mgy$
- $W_{\text{gravity}} = PE_i - PE_f$
- Units of Potential Energy are the same as those of Work and Kinetic Energy





Example

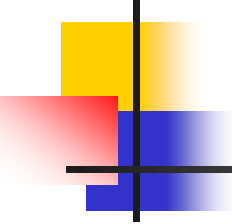
A 2kg bucket is 4.00m high, find gravitational potential energy

solution

$$PE = mgh$$

$$PE = (2.00 \text{ kg})(9.80 \text{ m/s}^2)(4.00 \text{ m})$$

$$PE = 78.4 \text{ J}$$

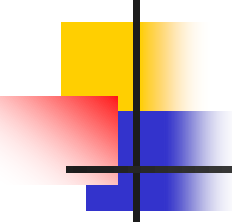


Work-Energy Theorem, Extended

- The work-energy theorem can be extended to include potential energy:

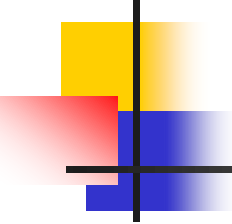
$$W_{nc} = (KE_f - KE_i) + (PE_f - PE_i)$$

- If other conservative forces are present, potential energy functions can be developed for them and their change in that potential energy added to the right side of the equation



Types of Forces

- There are two general kinds of forces
 - Conservative
 - Work and energy associated with the force can be recovered **e.g Gravity, Spring force, Electromagnetic forces**
 - Nonconservative
 - The forces are generally dissipative and work done against it cannot easily be recovered **e.g kinetic friction, air drag, propulsive forces**



Conservation of Mechanical Energy

- Conservation in general
 - To say a physical quantity is *conserved* is to say that the numerical value of the quantity remains constant throughout any physical process
- In Conservation of Energy, the total mechanical energy remains constant
 - *In any isolated system of objects interacting only through conservative forces, the total mechanical energy of the system remains constant.*



Mechanical energy is conserved when there is no non-conservative force.

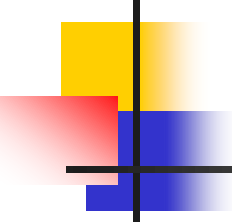
From the equation:

$$W_{nc} = \Delta KE + \Delta PE$$

Letting $W_{nc} = 0$, and putting the initial and final values to the left and the right of the equation respectively, we get the conservation of mechanical energy equation:

$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{1}{2}mv_i^2 + mgy_i = \frac{1}{2}mv_f^2 + mgy_f$$



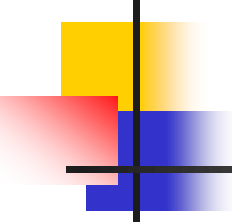
Conservation of Energy, cont.

- Total mechanical energy is the sum of the kinetic and potential energies in the system

$$E_i = E_f$$

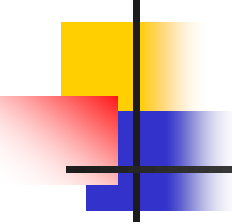
$$KE_i + PE_i = KE_f + PE_f$$

- Other types of potential energy functions can be added to modify this equation



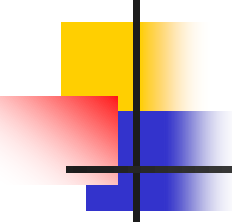
Problem Solving with Conservation of Energy

- Define the system
- Select the location of zero gravitational potential energy
 - Do *not* change this location while solving the problem
- Identify two points the object of interest moves between
 - One point should be where information is given
 - The other point should be where you want to find out something



Problem Solving, cont

- Verify that only conservative forces are present
- Apply the conservation of energy equation to the system
 - Immediately substitute zero values, then do the algebra before substituting the other values
- Solve for the unknown(s)



Example

A child of mass m rides on an irregularly curved slide of height $h=2.00\text{m}$. The child starts from rest at the top.

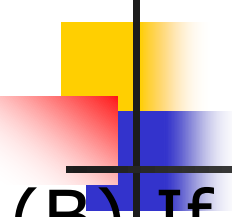
(A) Determine his speed at the bottom, assuming no friction is present.

solution

$$K_f + U_f = K_i + U_i$$

$$\frac{1}{2}mv_f^2 + 0 = 0 + mgh$$

$$v = \sqrt{2gh} = \sqrt{2(9.80 \text{ m/s}^2)(2.00 \text{ m})} = 6.26 \text{ m/s}$$



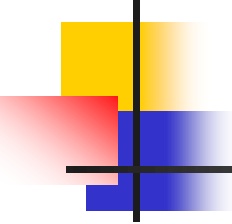
Cont...

(B) If a force of kinetic friction acts on the child, how much mechanical energy does the system lose? Assume that $v_f = 3.00 \text{ m/s}$ and $m = 20.0 \text{ kg}$.

Solution

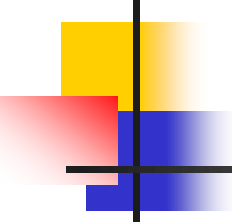
$$\begin{aligned}\Delta E_{\text{mech}} &= (K_f + U_f) - (K_i + U_i) \\ &= \left(\frac{1}{2} m v_f^2 + 0\right) - (0 + mgh) = \frac{1}{2} m v_f^2 - mgh \\ &= \frac{1}{2} (20.0 \text{ kg})(3.00 \text{ m/s})^2 \\ &\quad - (20.0 \text{ kg})(9.80 \text{ m/s}^2)(2.00 \text{ m}) \\ &= -302 \text{ J}\end{aligned}$$

- *Negative indicate reduction in mechanical energy*



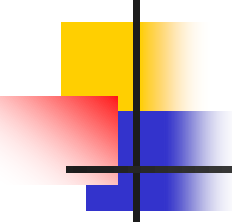
Work-Energy With Nonconservative Forces

- If nonconservative forces are present, then the full Work-Energy Theorem must be used instead of the equation for Conservation of Energy
- Often techniques from previous chapters will need to be employed



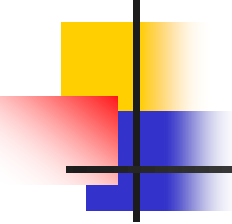
Potential Energy Stored in a Spring

- Involves the *spring constant*, k
- Hooke's Law gives the force
 - $F = -kx$
 - F is the restoring force
 - F is in the opposite direction of x
 - k depends on how the spring was formed, the material it is made from, thickness of the wire, etc.



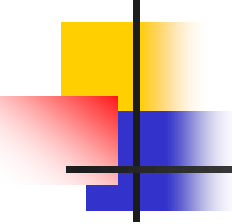
Potential Energy in a Spring

- Elastic Potential Energy
 - related to the work required to compress a spring from its equilibrium position to some final, arbitrary, position x
 - $PE_s = \frac{1}{2}kx^2$



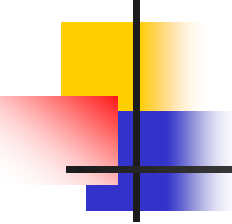
Work-Energy Theorem Including a Spring

- $W_{nc} = (KE_f - KE_i) + (PE_{gf} - PE_{gi}) + (PE_{sf} - PE_{si})$
 - PE_g is the gravitational potential energy
 - PE_s is the elastic potential energy associated with a spring
 - PE will now be used to denote the total potential energy of the system



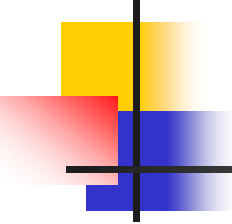
Conservation of Energy Including a Spring

- The PE of the spring is added to both sides of the conservation of energy equation
- $(KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f$
- The same problem-solving strategies apply



Nonconservative Forces with Energy Considerations

- When nonconservative forces are present, the total mechanical energy of the system is *not* constant
- The work done by all nonconservative forces acting on parts of a system equals the change in the mechanical energy of the system
 - $W_{nc} = \Delta Energy$



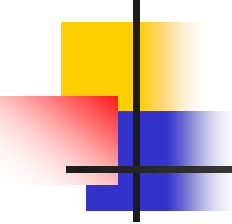
Nonconservative Forces and Energy

- In equation form:

$$W_{nc} = (KE_f - KE_i) + (PE_i - PE_f) \text{ or}$$

$$W_{nc} = (KE_f + PE_f) - (KE_i + PE_i)$$

- The energy can either cross a boundary or the energy is transformed into a form of non-mechanical energy such as thermal energy



Notes About Conservation of Energy

- We can neither create nor destroy energy
 - Another way of saying energy is conserved
 - If the total energy of the system does not remain constant, the energy must have crossed the boundary by some mechanism
 - Applies to areas other than physics



Example

A block having a mass of 0.80kg is given an initial velocity of 1.2m/s to the right and collides with a spring of negligible mass and force constant $k=50\text{N/m}$. Assuming the surface to be frictionless, calculate the maximum compression of the spring after the collision.

Solution

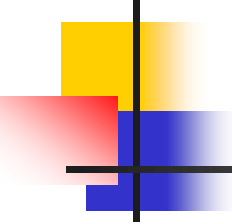
$$E_C = E_A$$

$$K_C + U_{sC} = K_A + U_{sA}$$

$$0 + \frac{1}{2} kx_{\max}^2 = \frac{1}{2} mv_A^2 + 0$$

$$x_{\max} = \sqrt{\frac{m}{k}} v_A = \sqrt{\frac{0.80 \text{ kg}}{50 \text{ N/m}}} (1.2 \text{ m/s})$$

$$= 0.15 \text{ m}$$



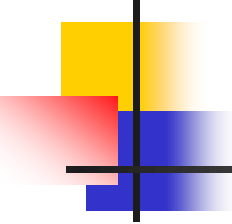
Power

- Often also interested in the *rate* at which the energy transfer takes place
- *Power* is defined as this rate of energy transfer

- $\bar{\dot{\phi}} = \frac{W}{t} = F\bar{v}$

- SI units are Watts (W)

- $W = \frac{J}{s} = \frac{kg \cdot m^2}{s^2}$

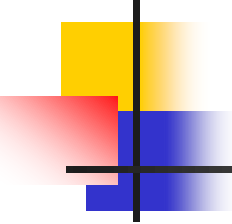


Power, cont.

- US Customary units are generally hp
 - Need a conversion factor

$$1 \text{ hp} = 550 \frac{\text{ft lb}}{\text{s}} = 746 \text{ W}$$

- Can define units of work or energy in terms of units of power:
 - kilowatt hours (kWh) are often used in electric bills
 - This is a unit of energy, not power

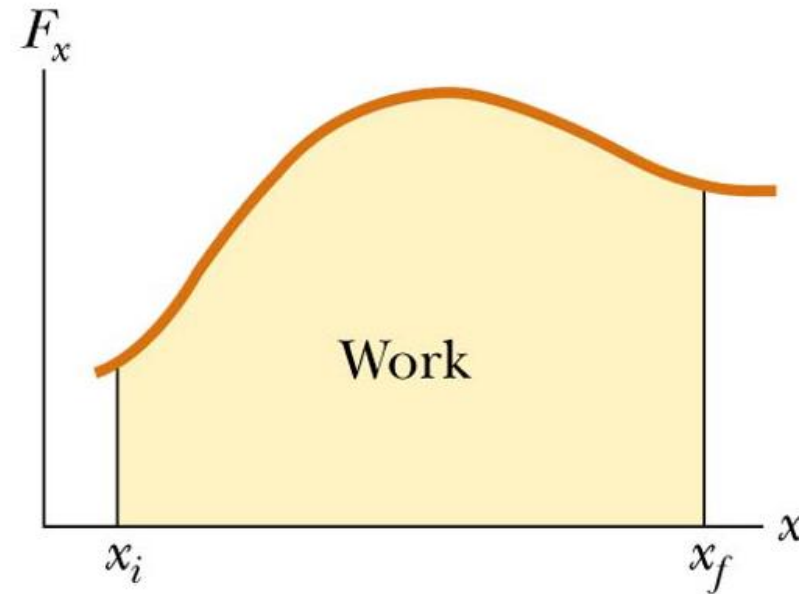


Center of Mass

- The point in the body at which all the mass may be considered to be concentrated
 - When using mechanical energy, the change in potential energy is related to the change in height of the center of mass

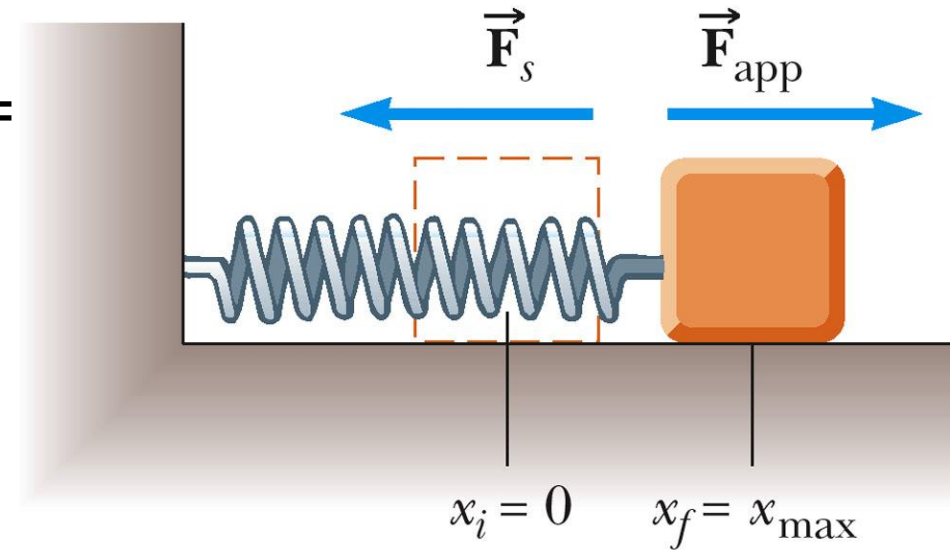
Work Done by Varying Forces

- The work done by a variable force acting on an object that undergoes a displacement is equal to the area under the graph of F versus x



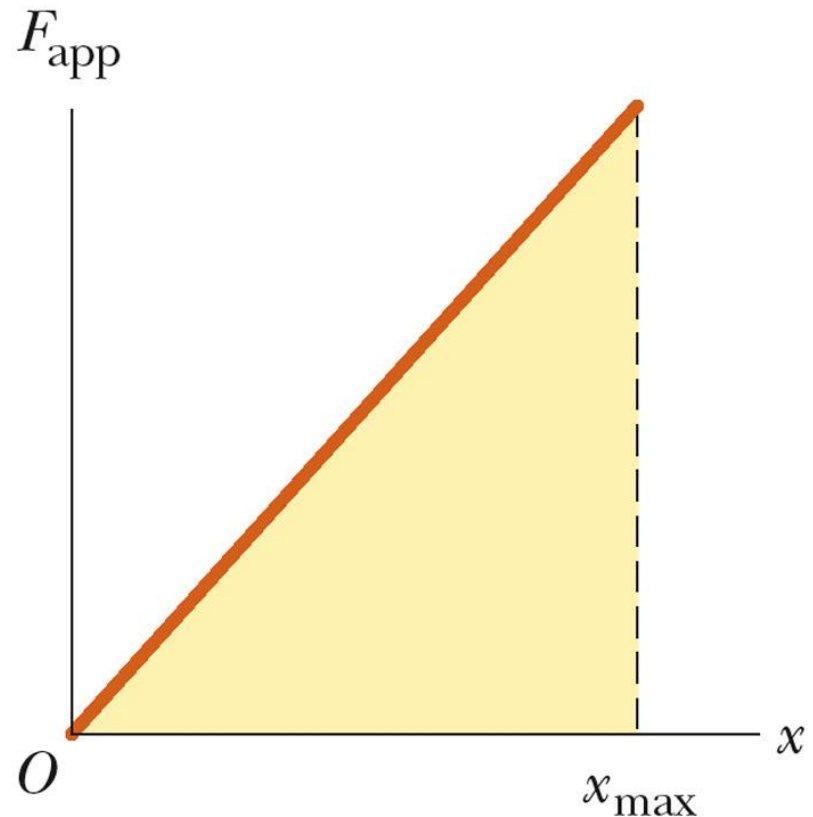
Spring Example

- Spring is slowly stretched from 0 to $x_{\max}$
- $\vec{F}_{\text{applied}} = -\vec{F}_{\text{restoring}} =$
- $W = \frac{1}{2}kx^2$



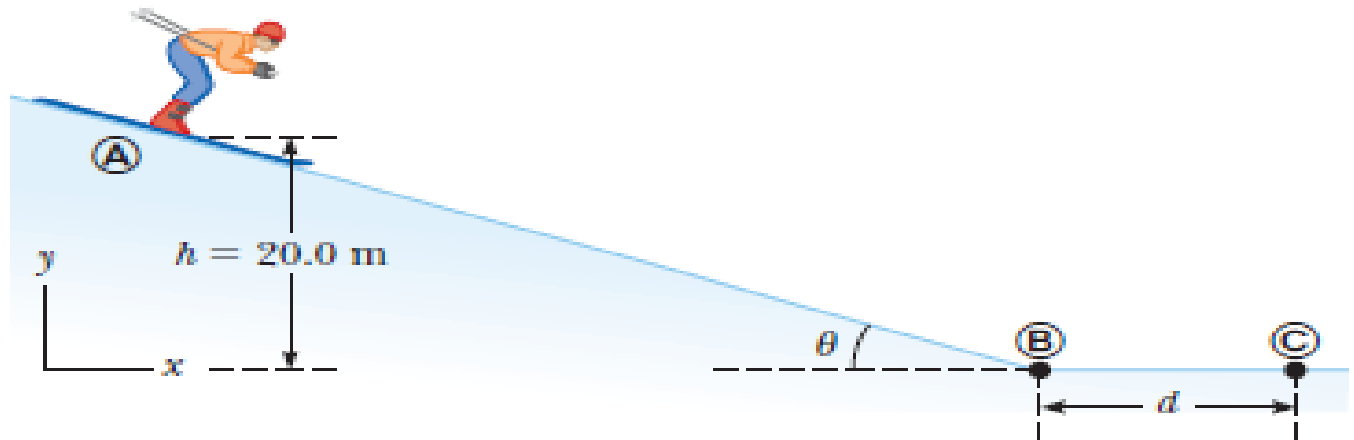
Spring Example, cont.

- The work is also equal to the area under the curve
- In this case, the “curve” is a triangle
- $A = \frac{1}{2} B h$ gives $W = \frac{1}{2} k x^2$



Example

A skier starts from rest at the top of a frictionless incline of height 20.0 m, as in Figure 5.19. At the bottom of the incline, the skier encounters a horizontal surface where the coefficient of kinetic friction between skis and snow is 0.210. **(a) Find the skier's speed at the bottom. (b) How far** does the skier travel on the horizontal surface before coming to rest? Neglect air resistance.



Example

A block with mass of 5.00 kg is attached to a horizontal spring with spring constant $4.00 \times 10^2 \text{ N/m}$, as in Figure 5.21. The surface the block rests upon is frictionless. If the block is pulled out to $x_i = 0.050 \text{ m}$ and released, **(a) find the speed** of the block when it first reaches the equilibrium point, **(b) find the speed when $x = 0.025 \text{ m}$** , and **(c) repeat part (a) if friction acts on the block, with coefficient of kinetic friction 0.150.**

